Lesson 7.5 Guided Notes

Session 1 · Plotting Points on the Coordinate Plane

I can plot a point with rational coordinates and name the quadrant it lies in. · 6.NS.C.6c

Word / Spanish	Meaning — complete the blank	Example
coordinate plane / plano de coordenadas	A grid made by a horizontal and a vertical _____.	x-axis across, y-axis up and down
ordered pair / par ordenado	Two numbers (x, y) that name one _____.	(−3, 4)
origin / origen	The point _____ where the two axes cross.	start every plot here
quadrant / cuadrante	One of the _____ sections of the plane.	Quadrant II is upper left

Key idea

The x-coordinate tells you to move _____. The y-coordinate tells you to move up or down.

Signs name the quadrant: (−, +) is Quadrant _____ and (−, −) is Quadrant III.

Worked example

Plot the point (−5, −4). Name its quadrant.

Start at the origin, (0, 0).

Move 5 units _____, because x = −5.

Move 4 units _____, because y = −4.

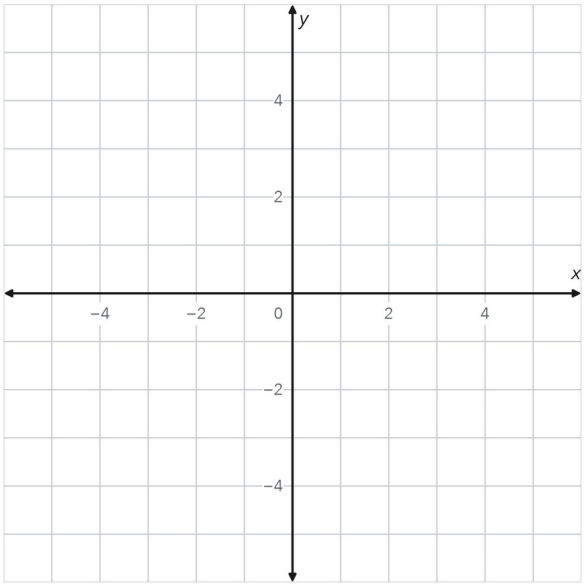
Answer: (−, −) means Quadrant _____, the lower left.

You try

1. Name the quadrant of each point.
a. (6, −2) _____
b. (−1.5, 3) _____

_____

2. Plot and label A(−4, 2) and B(3, −5).



_____

Sum it up: To plot (x, y), I start at ___, then I ___.

_____

Session 2 · Reflecting Points Across Axes

I can reflect a point across the x-axis or the y-axis and name the new coordinates. · 6.NS.C.6b

Word / Spanish	Meaning — complete the blank	Example
reflection / reflexión	A _____ that makes a mirror image across a line.	(4, 1) becomes (4, −1) across the x-axis
x-axis / eje x	The _____ number line through the origin.	runs left and right
y-axis / eje y	The _____ number line through the origin.	runs up and down

Key idea

Across the x-axis, the _____-coordinate changes sign.

Across the y-axis, the _____-coordinate changes sign.

Worked example

Reflect (7, −2) across the x-axis. Then reflect (7, −2) across the y-axis.

Across the x-axis: keep x = 7 and change −2 to _____. Image: (7, 2).

Across the y-axis: keep y = −2 and change 7 to _____.

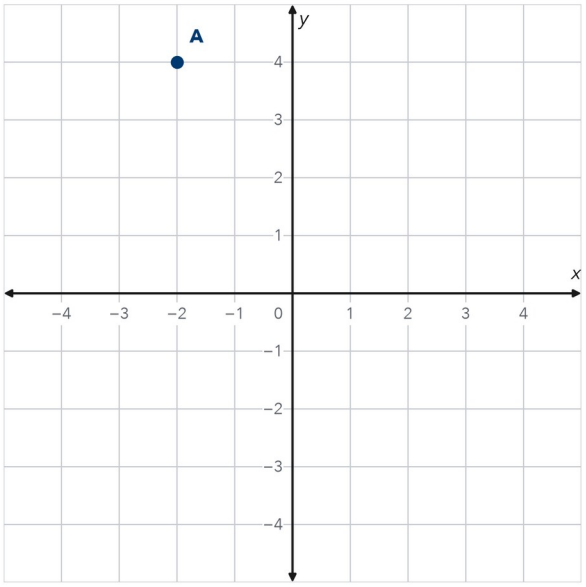
Answer: The images are (7, 2) and _____.

You try

1. Write each image.
a. (3, −5) across the x-axis _____
b. (−6, 1) across the y-axis _____

_____

2. Point A is at (−2, 4). Plot A and its reflection B across the x-axis.



_____

Sum it up: When I reflect across the y-axis, ___ changes because ___.

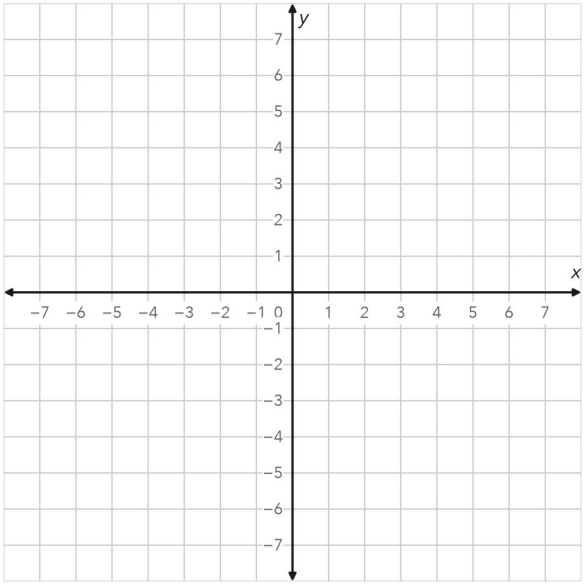
_____

Session 1 · Model, reason, and explain

Plotting Points on the Coordinate Plane · 6.NS.C.6c

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Plot (−2,3), (0,−3), and (0,0). Name the quadrant where appropriate and explain why axis points have no quadrant.



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Extend your reasoning

Create a contrasting example that exposes the misconception: assigns an axis point to a quadrant.. Explain what changes and what stays true.

_____

_____

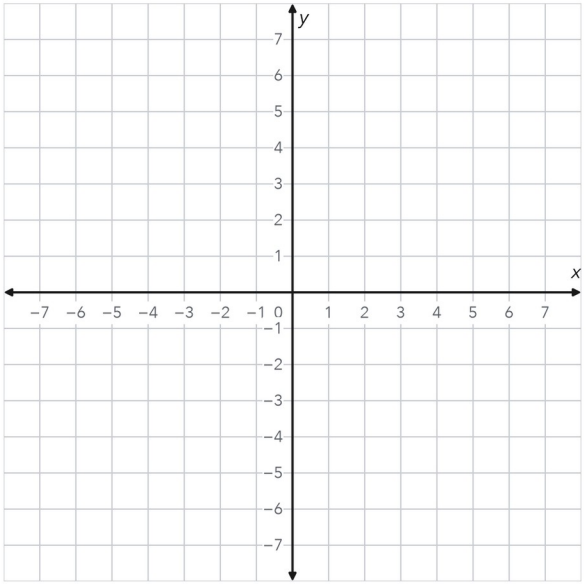
Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Reflecting Points Across Axes · 6.NS.C.6b

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Reflect (3,−2) across each axis separately, then reflect it across the x-axis followed by the y-axis. Also reflect (0,4) across the y-axis.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: applies an independent reflection to the wrong starting image or changes both coordinates in one reflection.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?